# Belief-State Geometry in Transformers: Reproduction and Unsupervised Recovery via Observable Operator Models

## 1 Introduction

Shai et al.[1] show that a transformer trained on sequences from a hidden data-generating process linearly represents, in its residual stream, the geometry of *belief states*: the Bayesian posterior distributions over the generator's hidden states. The geometry is characterised by the *mixed-state presentation* (MSP): each history $w$ induces a belief $b(w)$ over the hidden states, and the set of reachable beliefs forms a characteristic structure in the probability simplex over the hidden states (for classical data-generating processes)[2].

This work reproduces the supervised results and contributes an *unsupervised* estimator that recovers the belief-state geometry from activations and the network's own next-token probabilities alone, without ground-truth belief labels or knowledge of the generating process, using observable operator models (OOMs). We present supervised and unsupervised belief state recovery on two processes; *Mess3*, a 3-state dense process with Sierpinski fractal geometry, and *RRXOR*, a 5-state synchronising process with 36 discrete belief states.

Both processes are trained using the paper's architecture ($d_{\text{model}}$=64, 4 layers, 1 head, $d_{\text{mlp}}$=256, ReLU, LayerNorm, SGD) for $10^6$ steps. Throughout, we use prefix deduplication: since causal attention ensures identical contexts produce identical activations, each unique prefix appears once (weighted by its total frequency) rather than at every position. We weight all regressions by prefix probability $P(w)$, following [2], to give priority to the model's learned geometry in high-probability regions.

A limitation of the supervised approach is requiring knowledge of the data-generating process, which prevents us from applying this approach to large language models. If we could recover the belief geometry in an unsupervised way this would address this limitation. If transformers truly represent belief-state geometry linearly in their residual streams, then this geometry should be recoverable without labels or access to the generating process. Success on these controlled experiments would suggest that the approach could extend to large language models.

## 2 Results: Supervised vs. Unsupervised Recovery

### 2.1 The unsupervised estimator

The estimator recovers the belief subspace from activations and the network's own softmax, using the *observability matrix* of an observable operator model (OOM) [3], but estimated in activation space. We refer to this estimator as the spectral-OOM.

**Linearising the belief update.** The network encodes belief affinely, $a(w) = b(w)\Psi + c$, where $\Psi$ is the linear embedding and $c$ is a constant offset. Because every regression below is fit with an intercept (equivalently, on $P(w)$-centered activations), the offset $c$ is absorbed and the operators are estimated on the linear part $b(w)\Psi$. The Bayesian update is projective (Eq. (1)),

$$b(wx) = \frac{b(w)T^x}{b(w)T^x\mathbf{1}}, \qquad P(x|w) = b(w)T^x\mathbf{1}. \tag{1}$$

a homography: a linear map $b(w) \mapsto b(w)T^x$ followed by a normalisation that divides by a linear functional of $b(w)$. The ratio is non-linear in $b(w)$, and hence in $a(w)$, so no fixed matrix relates consecutive activations across all prefixes $w$.

However, observe that the *normalising denominator is exactly the next-token probability* $P(x|w) = b(w)T^x\mathbf{1}$, which is the network's softmax output. Multiplying both sides by $P(x|w)$ removes the non-linearity, giving us

$$b(wx)\,P(x|w) = b(w)\,T^x, \tag{2}$$

whose right-hand side is now linear in $b(w)$. Right-multiplying (2) by $\Psi$ rewrites it in activation coordinates. On the left, $P(x|w)$ is a scalar and $b(wx)\Psi = a(wx)$, giving $P(x|w)\,a(wx)$. On the right, inserting $\Psi\Psi^+$ (the identity on the belief row-space when $\Psi$ has full row rank $d$, i.e. the encoding is injective) regroups $b(w)\Psi$ into $a(w)$:

$$b(w)\,T^x\,\Psi = b(w)\,\big(\Psi\Psi^+\big)\,T^x\,\Psi = \big(b(w)\Psi\big)\big(\Psi^+T^x\Psi\big) = a(w)\,A_x, \qquad A_x := \Psi^+T^x\Psi.$$

Equating the two sides yields a single fixed operator $A_x$ relating every parent activation $a(w)$ to its rescaled child $P(x|w)\,a(wx)$ (where child activation means the appended-token activation $a(wx)$)

$$a(w)\,A_x \;\approx\; P(x|w)\,a(wx). \tag{3}$$

The relation is approximate because $\Psi$ is unknown, belief is only approximately encoded, and $A_x$ are estimated from finite data. The map $A_x = \Psi^+T^x\Psi$ is simply the hidden-state operator $T^x$ expressed in activation coordinates. Crucially, the rescaling by the network's own $P(x|w)$ is what absorbs the prefix-dependent normaliser that made the update projective, leaving a relation that is *linear* in the sense that one matrix $A_x$ holds for all $w$. Each $A_x$ is thus estimable by a single ($P(w)$-weighted) linear regression over transition pairs $\big(a(w),\,a(wx)\big)$ rescaled by $P(x|w)$, no supervised targets are required.

**Recovering the subspace.** The operators $A_x$ of Eq. (3) advance the state *within* activation space; we also need a *readout* from a state to what it predicts. Let $C$ be the ($P(w)$-weighted) ridge regression of activations $a(w)$ onto the softmax row $P(\cdot\,|\,w)$, the one-step observable. It is linear in belief because $P(x|w) = b(w)T^x\mathbf{1}$ (Eq. (1)) is a linear functional of $b(w)$, so the full softmax row is a linear image of $b(w)$ and hence of $a(w)$. We fit $C$ as a linear map rather than reuse the network's own activation-to-softmax map because the latter is non-linear (the softmax) and so cannot be composed into the matrix products below. $C$ is the best linear readout to the next-token probability simplex, and the SVD acts on matrices. Both $C$ and $\{A_x\}$ are fit by the same regression from one pass over the prefix tree, independently and in either order. Their only targets are the network's softmax and (rescaled) child activations.

The pair $(C, \{A_x\})$ is the activation-space analogue of the output map and dynamics of a linear system. $C$ says what is seen now, $A_x$ says how the state evolves, so $A_xC$ is what is seen one step ahead, $A_xA_yC$ two steps ahead, and so on. Composing the $A_x$ preserves linearity in $b(w)$, so every such multi-step observable is again linear in belief. Stacking them up to a horizon $L$ gives the *observability matrix*

$$\mathcal{O} = \big[\, C \;\big|\; (A_x C)_x \;\big|\; (A_x A_y C)_{x,y} \;\big|\; \cdots \;\big|\; (A_{x_1} \cdots A_{x_L} C)_{x_1,\dots,x_L} \,\big], \tag{4}$$

indexed over all words up to length $L$. This is the (word-indexed) control-theoretic observability matrix of the estimated OOM; $L$ is taken large enough that $\mathcal{O}$ reaches its full rank, i.e. enough future to make the hidden states distinguishable.

The column space of $\mathcal{O}$ is the image of belief space under the encoding, so it spans exactly the belief-carrying directions of activation space, the *belief subspace*, isolating it from the rest of the residual stream. Its dimension, the numerical rank of $\mathcal{O}$, is the model order. For an observable (minimal) process this equals the number of hidden states, while a weakly observable process exposes only its observable part and can rank below the true state count (see Section 2.3, RRXOR). An SVD $\mathcal{O} = U\Sigma V^\top$ then does two things: the singular-value spectrum locates the effective rank $d$ (the elbow giving the model order), and the leading left singular vectors $B = U_{:,1:d}$ furnish an orthonormal basis for the belief subspace, which we use to project activations. Recovering a state subspace via the SVD of a matrix of observable statistics is the standard spectral-learning route [5, 4]. There they factor the past$\times$future Hankel matrix, whereas here we build its observability factor directly from learned operators in activation space. The procedure uses only activations, softmax, and the prefix tree.

### 2.2 Mess3

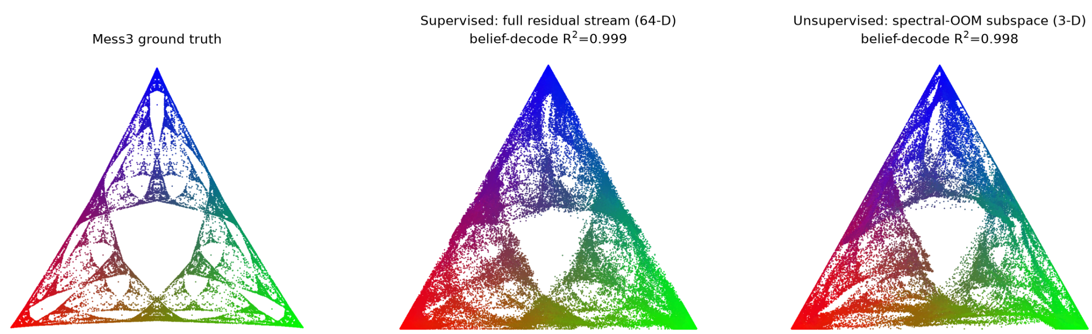


Figure 1: **Mess3: ground truth, supervised, and unsupervised.** Left: ground-truth belief geometry. Centre: supervised decode from full 64-D residual stream ($R^2 = 0.999$). Right: unsupervised spectral-OOM subspace ($R^2 = 0.998$). The estimator recovers the fractal nearly identically.

For the Mess3 process, unsupervised recovery is nearly perfect. The observability spectrum drops cleanly at $d = 3$, and the recovered 3-D subspace decodes belief at $R^2 = 0.998$, almost as good as supervised. The geometry emerges similarly during training (Figure 2) and survives identical controls: both cross-validate on held-out prefixes and collapse under belief-label (supervised) or softmax (unsupervised) shuffle (Figure 3). The 30/70 split is a slight change from the 20/80 split in the original because the original split provides too little data for unsupervised learning; its training unit is a transition (parent + all children), while supervised trains on single prefixes.

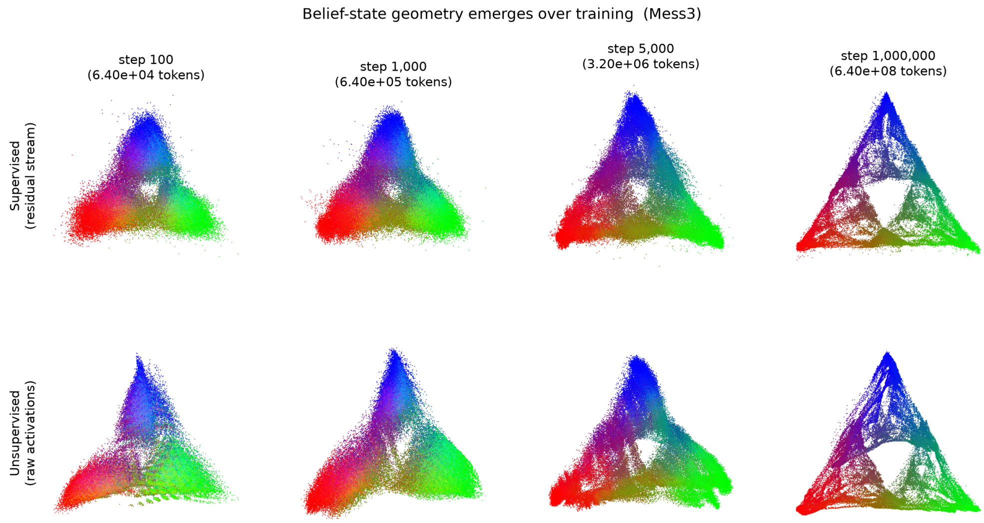


Figure 2: **Mess3 emergence over training.** Top: supervised decode at steps $100, 1000, 5000, 10^6$. Bottom: spectral-OOM subspace activations (affine-aligned for display). Both show identical emergence trajectory.

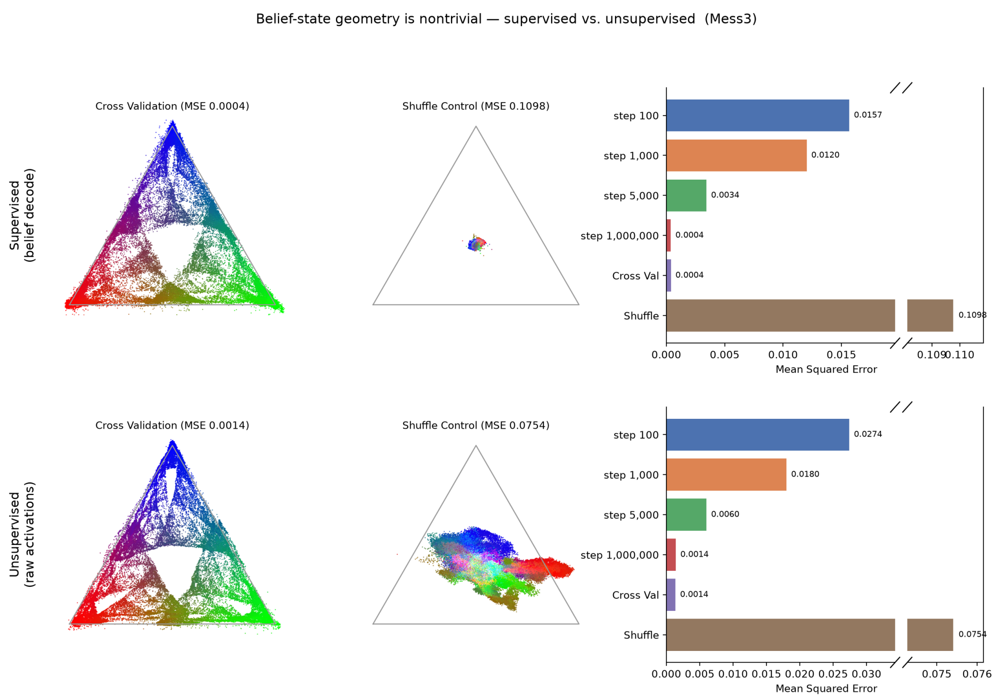


Figure 3: **Mess3 controls: cross-validation and shuffle.** Top: supervised. Bottom: unsupervised. Both recover the fractal on held-out prefixes (CV) and collapse dramatically under shuffle control.

## 2.3 RRXOR

For RRXOR the gap to supervised recovery is substantial. The unsupervised subspace recovers the diamond geometry but more diffusely (Figure 4). We report two complementary scores: per-position $R^2$ (full point cloud) and state-level $R^2$ (36 belief state centers-of-mass). The unsupervised subspace gives per-position $R^2 = 0.65$ (supervised 0.91) and state-level $R^2 = 0.82$ (supervised 1.00). The $d = 5$ dimension was fixed *a priori* from the known RRXOR state count, since the singular spectrum showed no clean elbow for this process.

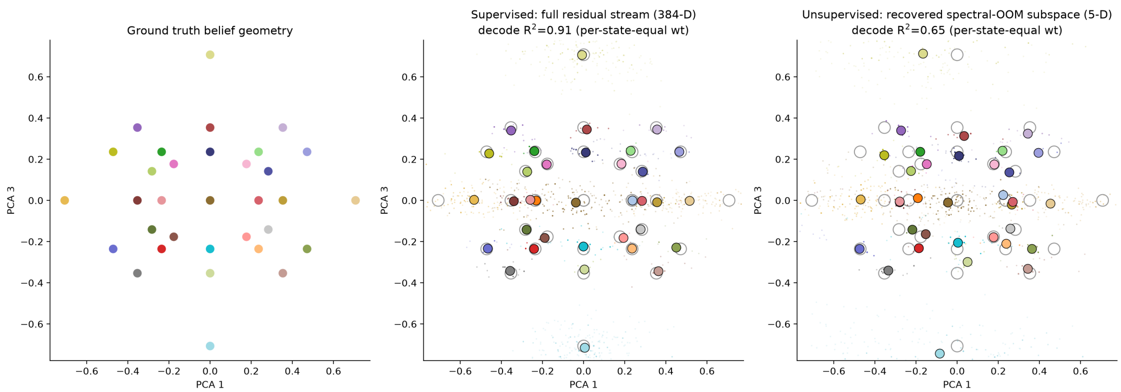


Figure 4: **RRXOR: ground truth, supervised, and unsupervised.** Left: ground-truth 36 belief states (PC1/PC3 plane, display only). Centre: supervised decode (per-position $R^2 = 0.91$). Right: recovered subspace (per-position $R^2 = 0.65$, state-level $R^2 = 0.82$). The gap reflects weak observability.

This gap is structural, not an artefact of estimator quality. The root cause is rank deficiency in the observability matrix: RRXOR's belief directions produce only faint singular values in the predictive observables (the multi-step version of the 0.33 next-token limitation), making them difficult to recover unsupervised. We tested observability at horizons $L = 3$ through $L = 15$ with $P(w)$-weighting on the full length-10 enumeration. The state-level geometry $R^2$ at $d = 5$ shows that longer horizons do not improve recovery, best performance is at $L = 3$ ($R^2 = 0.815$). This confirms that observability is maximized at the shortest tested horizon.

The layerwise analysis and distance preservation are recovered in the supervised case (Figure 5). Belief is spread across all layers (no single layer encodes it well), and is preserved in distances better than next-token. Notably, the unsupervised representation reverses this (Figure 6). Next-token distances ($R^2 \approx 0.68$) are preserved better than belief ($R^2 \approx 0.50$). A possible explanation for this might be that the spectral-OOM subspace is too biased towards the next-token observable $C$. Since RRXOR's belief geometry is not strongly correlated with next-token structure, this method may not be well-poised to recover the true belief geometry.

# 3 Conclusion

The belief-state geometry can be recovered unsupervised from activations and softmax alone, nearly perfectly for Mess3 ($R^2 \approx 0.998$), but only partially for RRXOR (state-level $R^2 \approx 0.82$). The spectral-OOM anchors to next-token predictions (the softmax observable), but RRXOR's belief states exhibit significant next-token degeneracy, i.e. many geometrically distinct states share similar next-token distributions ($R^2$=0.31 from the original paper). It is unclear whether this issue with the spectral-OOM estimator can be fully addressed, as there are no obvious alternative observables satisfying the required properties (linearity in belief, empirical accessibility). A longer context window could mitigate this by providing more data per horizon depth, reducing compounding errors in deep recursion. Success on Mess3 demonstrates the method's viability for processes where belief geometry aligns with observable structure. A natural next experiment is a richly-sampled quantum or post-quantum process where this alignment should hold more generally.

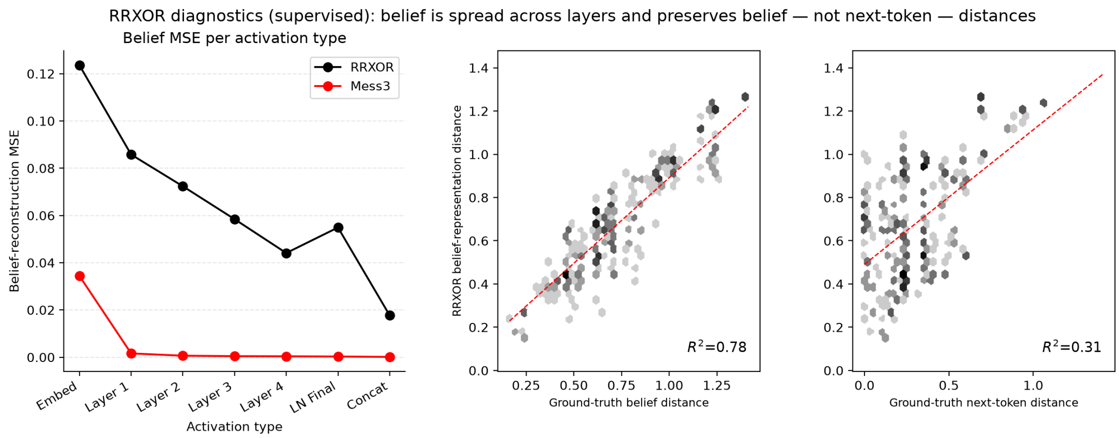


Figure 5: **RRXOR diagnostics: supervised.** Left: per-activation belief-reconstruction MSE. Centre/right: belief and next-token distance preservation via hexbin.

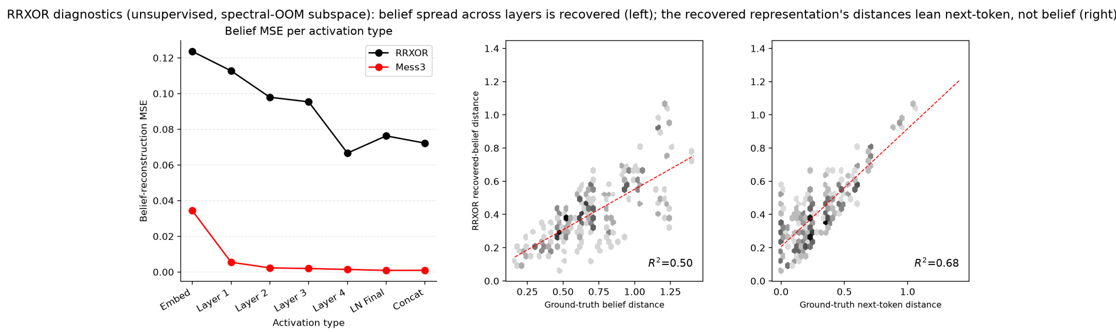


Figure 6: **RRXOR diagnostics: unsupervised.** Same metrics as Figure 5, but from the recovered spectral-OOM subspace. Belief spread is recovered, but next-token distances dominate.

# A Observable Operator Models (OOMs): Spectral-OOM Estimator Details

## A.1 The spectral-OOM estimator on transformer activations

From Section 2.1, the network encodes belief affinely: $a(w) = b(w)\Psi + c$. Working in $P(w)$-centered coordinates (the offset $c$ is absorbed by the regression intercept), the projective update (1) is nonlinear in $a$, but its normaliser is precisely the softmax $P(x|w) = b(w)T^x\mathbf{1}$. Multiplying both sides by the normaliser gives $b(wx)\,P(x|w) = b(w)T^x$; right-multiplying by $\Psi$ linearises the update:

$$a(w)\,A_x \approx P(x|w)\,a(wx), \qquad A_x = \Psi^+ T^x \Psi. \tag{5}$$

**Building the observability matrix.** Let $C$ be the ridge regression of activations onto the softmax (the one-step observable), and $\{G_x\}$ the estimated transition operators (ridge regressions of activations onto $P(x|w)$·child-activations, i.e. the empirical version of $A_x$). All regressions are $P(w)$-weighted to bias the estimator toward high-probability prefixes. Because each $G_x$ is linear in belief and composition preserves linearity, every block below is linear in $b(w)$. Stacking multi-step observables to horizon 3 gives the observability matrix

$$\mathcal{O} = \left[\, C \;\middle|\; (G_x C)_x \;\middle|\; (G_x G_y C)_{x,y} \;\middle|\; (G_x G_y G_z C)_{x,y,z} \,\right], \tag{6}$$

indexed over all words up to length 3. This is the (word-indexed) control-theoretic observability matrix of the estimated OOM: with a single-symbol alphabet it reduces to the classical Kalman form $[\, C \mid CA \mid CA^2 \mid \dots ]$, and the token-indexed family $\{G_x\}$ is the natural OOM/PSR generalisation. An SVD $\mathcal{O} = U\Sigma V^\top$ yields the orthonormal basis $B = U_{:,1:d}$, with the numerical rank $d$ giving the model order.

**What is and isn't unsupervised.** Subspace choice, operator fit, and one-step $R^2$ are fully target-free (activations + softmax only). Order selection is target-free where the spectrum has a clean elbow (Mess3, $d = 3$); for the weakly observable RRXOR there is no elbow, and $d = 5$ is set a priori from the known state count (Section 2.3). The final placement onto the simplex and numerical comparisons use ground-truth beliefs, but only for display and scoring, not for fitting.

# B Operator Rollouts

A stronger test of the subspace is the generative *operator rollout*. The activation-space operators $A_x$ of Eq. (3) were used only to build the observability matrix and locate the subspace; they are not iterated. The rollout instead asks whether a self-contained dynamical system fitted *inside* the recovered subspace can regenerate the entire geometry from the root alone.

Three steps produce it. (i) *Operator-consistency refit*: the recovered basis $B$ is good for decoding but is not exactly closed under the rescaled-transition relation, so operators fitted in it drift when iterated; we refine $B$ by alternating least squares (ALS) into an operator-consistent basis $B_{\text{op}}$, alternating between fitting operators in the current basis and rotating the basis toward one those operators map into itself.

## B.1 Results

**Mess3:** the rollout reproduces the full fractal at $R^2 \approx 1.00$ from both supervised and unsupervised subspaces (Figures 7 and 8). Mess3's true operators are diagonalisable strict contractions; the fractal is their attractor, and the rollout self-corrects.

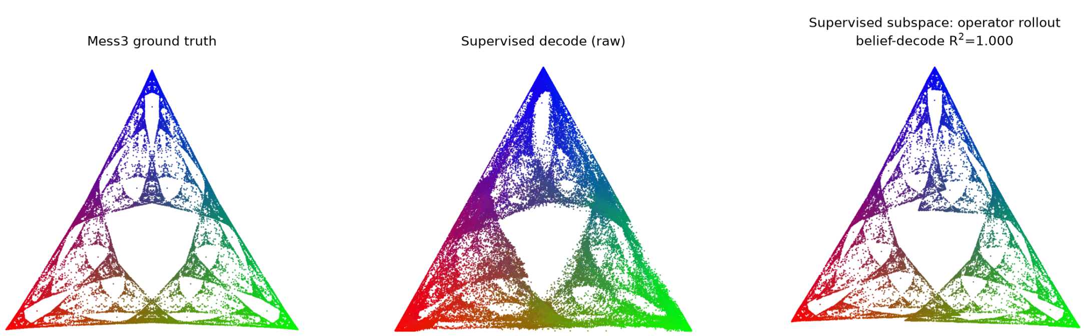


Figure 7: **Mess3 supervised subspace.** Ground truth | supervised decode | operator rollout ($R^2 = 1.00$).

**RRXOR:** the rollout collapses, failing to reach the outer apexes from both supervised and unsupervised subspaces. This is intrinsic to RRXOR's operator algebra: the true belief-update operators

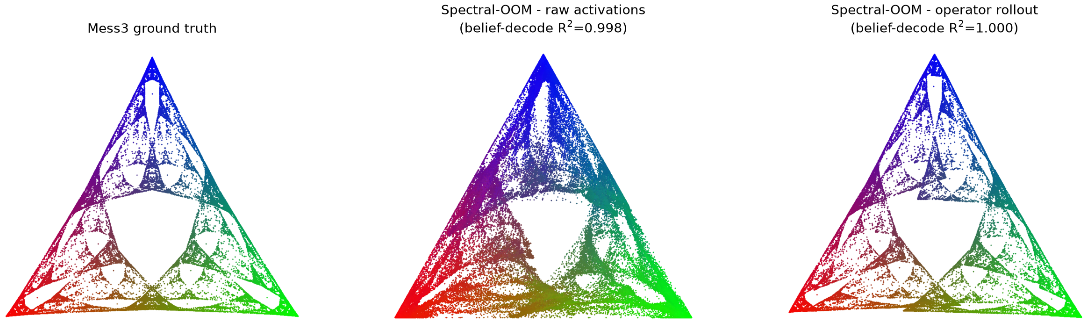


Figure 8: **Mess3 spectral-OOM subspace.** Ground truth | recovered subspace (raw) | operator rollout via $\widehat{A}_x$ ($R^2 = 1.00$).

are non-diagonalisable (nilpotent/defective), so the belief states are transients of a defective semi-group, not an attractor set. Hence for RRXOR the raw subspace (main text), not the rollout, is the trustworthy representation.

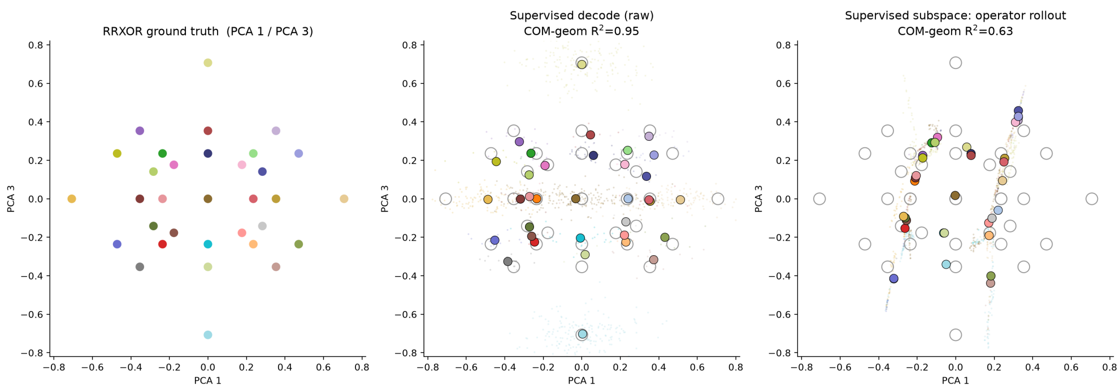


Figure 9: **RRXOR supervised subspace.** Ground truth | supervised decode (COM-geom 0.95) | operator rollout (COM-geom 0.63; extreme states not reached).

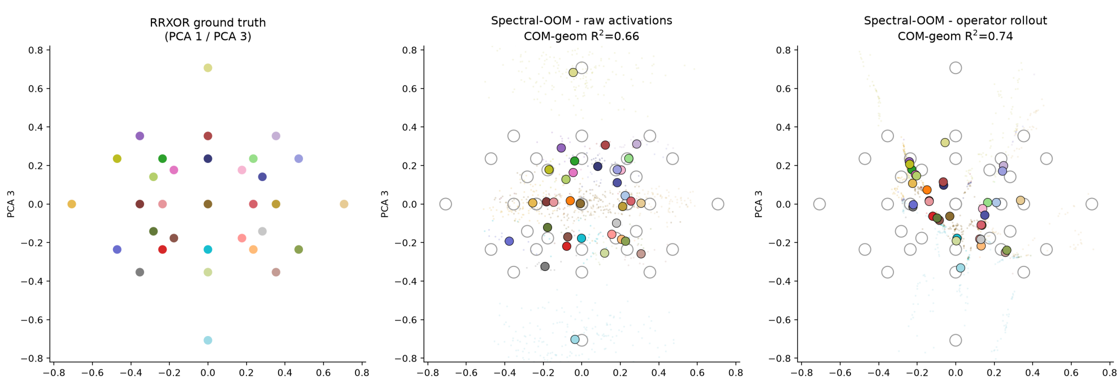


Figure 10: **RRXOR spectral-OOM subspace.** Ground truth | recovered subspace (raw, state-level $R^2 = 0.66$) | operator rollout via $\widehat{A}_x$ (state-level $R^2 = 0.74$; extreme states central).

# References


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